BRAIN SIGNALS
AI
ROBOTICS
A SAFER TOMORROW

# CAI

COGNITIVE ASTRONAUT INTERFACE

CONTROL BEYOND TOUCH



HUMAN MIND × ARTIFICIAL INTELLIGENCE
FOR A SAFER TOMORROW

## ABOUT CAI

BRIDGING MINDS AND MACHINES BEYOND EARTH

The Cognitive Astronaut Interface (CAI) is a brain–computer interface that uses EEG signals to control a robotic assistant.

CAI allows astronauts to perform tasks hands-free using their focus, blinks and attention — helping them work more efficiently and safely in space.

It also continuously monitors brain activity to detect abnormal signals or lack of response, triggering an automatic safety protocol in emergencies.

## WHY IT MATTERS

**INCREASES SAFETY**
Detects early signs of distress and triggers emergency response.

**IMPROVES EFFICIENCY**
Hands-free control for faster and easier task completion.

**SUPPORTS THE FUTURE**
Helps make long-duration space missions safer and more independent.

“A SAFER TOMORROW STARTS WITH A BRIGHTER MIND”

## KEY FEATURES

INTELLIGENCE IN EVERY SIGNAL

**BRAIN SIGNAL INPUT**
EEG headset captures blinking, focus and attention in real time.

**AI SIGNAL PROCESSING**
Python processes and interprets EEG signals into commands.

**ROBOTIC CONTROL**
Commands are sent to a robotic assistant for movement and tasks.

**EMERGENCY DETECTION**
Abnormal signals or no response triggers automatic safety actions.

## OUR TECHNOLOGY

EEG Headset
Arduino Controller
Python AI Processing
Robotic Assistant

EXPLORING MORE
WITH A BRIGHTER MIND

## OUR TEAM

FOUR MINDS. A SAFER TOMORROW.

**MARIAM SAAD**
Team Leader

**SALAMA JAMAL**
Team Member

**SHAIKHA ALI**
Team Member

**MEERA ABDULHAKEEM**
Team Member

DIFFERENT MINDS
SAME MISSION
A SAFER TOMORROW

CONTROL BEYOND TOUCH